

Assignment 4

a. Based on accuracy which dimensionality reduction method, PCA, simulate annealing, or the genetic algorithm worked the best?

The Genetic Algorithm (GA) performed best with accuracy = 0.9667

b. For each of the two other methods, explain why you think it did not perform as well as the best one.

PCA: Chose only z_1 , which preserved variance but not class separation.

Simulated Annealing (SA): Couldn't escape local optima; its subset ['petal-width', 'z2'] missed key features.

c. Did the best dimensionality reduction method produce a better accuracy than using none (i.e. the results of Part 1)? Explain possible reasons why it did or did not.

GA outperformed the original features ($0.9667 > 0.9467$) by removing redundancy and combining stronger transformed features.

d. Did Part 2 produce the same set of best features as Part 3? Explain possible reasons why it did or did not.

Different results (z_1 vs ['petal-width', 'z2']), PCA is unsupervised, SA is accuracy-driven.

e. Did Part 2 produce the same set of best features as Part 4? Explain possible reasons why it did or did not.

Different results (z_1 vs z_1, z_3, z_4), GA combined multiple PCA components for better accuracy

f. Did Part 3 produce the same set of best features as Part 4? Explain possible reasons why it did or did not.

Different results (['petal-width', 'z2'] vs [z_1, z_3, z_4]), GA explores a population of solutions and found a more optimal subset.

Part 1: Original 4 features

Confusion matrix:

```
[ 50  0  0]
```

```
[  0 47  3]
```

```
[  0  5 45]
```

Accuracy: 0.9467

Features used: ['sepal-length', 'sepal-width', 'petal-length', 'petal-width']

Part 2: PCA transform (z1..z4)

Eigenvalues (lambda_i):

z1: 4.228242

z2: 0.242671

z3: 0.078210

z4: 0.023835

Eigenvectors matrix (rows = original features; columns = components z1..z4):

orig_feat_1: +0.361387, +0.656589, -0.582030, +0.315487

orig_feat_2: -0.084523, +0.730161, +0.597911, -0.319723

orig_feat_3: +0.856671, -0.173373, +0.076236, -0.479839

orig_feat_4: +0.358289, -0.075481, +0.545831, +0.753657

PoV (proportion of variance) per component:

z1: 0.924619

z2: 0.053066

z3: 0.017103

z4: 0.005212

Cumulative PoV: [0.92461872 0.97768521 0.99478782 1.]

Selected components (cumulative PoV > 0.90): ['z1'] (k = 1)

Part 2 Confusion matrix (using selected z components):

```
[ 50  0  0]
```

```
[  0 43  7]
```

```
[  0  6 44]
```

Part 2 Accuracy: 0.9133

Part 3: Simulated Annealing

Initial subset: ['sepal-length', 'sepal-width', 'petal-width'] | Accuracy: 0.9200
Iter 001: Subset=['sepal-length', 'petal-width'] | Accuracy=0.9067 | Pr[accept]=0.986622 | RandomUniform=0.487857 | Status=Accepted
Iter 002: Subset=['sepal-length', 'petal-width', 'z1', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
Iter 003: Subset=['sepal-length', 'petal-width', 'z1', 'z3', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.266331 | Status=Accepted
Iter 004: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z3', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
Iter 005: Subset=['sepal-width', 'petal-width', 'z1', 'z3'] | Accuracy=0.9400 | Pr[accept]=0.993007 | RandomUniform=0.025446 | Status=Accepted
Iter 006: Subset=['sepal-width', 'petal-width', 'z1'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.686484 | Status=Accepted
Iter 007: Subset=['sepal-length', 'sepal-width', 'petal-width'] | Accuracy=0.9200 | Pr[accept]=0.978724 | RandomUniform=0.221692 | Status=Accepted
Iter 008: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
Iter 009: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.230867 | Status=Accepted
Iter 010: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.952244 | Status=Accepted
*** Restart at iter 10: New random subset=['sepal-length', 'petal-length', 'petal-width', 'z2'] | Accuracy=0.9400 ***
Iter 011: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z2'] | Accuracy=0.9200 | Pr[accept]=0.977779 | RandomUniform=0.284159 | Status=Accepted
Iter 012: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
Iter 013: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2', 'z3', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
Iter 014: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2', 'z4'] | A

ccuracy=0.9400 | Pr[accept]=0.992278 | RandomUniform=0.414314 | Status=Accepted

Iter 015: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.548799 | Status=Accepted

Iter 016: Subset=['sepal-length', 'z1', 'z4'] | Accuracy=0.9600 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 017: Subset=['sepal-length', 'petal-length', 'z1', 'z4'] | Accuracy=0.9333 | Pr[accept]=0.968382 | RandomUniform=0.520938 | Status=Accepted

Iter 018: Subset=['sepal-length', 'petal-length', 'petal-width', 'z1', 'z2', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 019: Subset=['sepal-length', 'petal-length', 'petal-width', 'z1', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.043487 | Status=Accepted

Iter 020: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.502239 | Status=Accepted

*** Restart at iter 20: New random subset=['sepal-length', 'sepal-width', 'petal-length'] | Accuracy=0.9333 ***

Iter 021: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.783655 | Status=Accepted

Iter 022: Subset=['sepal-length', 'sepal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9600 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 023: Subset=['sepal-width', 'petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9400 | Pr[accept]=0.974360 | RandomUniform=0.870010 | Status=Accepted

Iter 024: Subset=['sepal-width', 'petal-width', 'z1', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.484925 | Status=Accepted

Iter 025: Subset=['sepal-width', 'petal-length', 'petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 026: Subset=['sepal-width', 'petal-length', 'petal-width', 'z1', 'z3'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=0.458147 | Status=Accepted

Iter 027: Subset=['sepal-width', 'petal-length', 'z1', 'z3'] | Accuracy=0.9400 |

Pr[accept]=0.990909 | RandomUniform=0.635371 | Status=Accepted

Iter 028: Subset=['petal-length', 'z1', 'z3'] | Accuracy=0.9333 | Pr[accept]=0.990783 | RandomUniform=0.798439 | Status=Accepted

Iter 029: Subset=['sepal-length', 'petal-length', 'z1'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 030: Subset=['sepal-length', 'sepal-width', 'petal-length', 'z1'] | Accuracy=0.9200 | Pr[accept]=0.971833 | RandomUniform=0.868078 | Status=Accepted

*** Restart at iter 30: New random subset=['sepal-width', 'petal-width', 'z2', 'z3'] | Accuracy=0.9267 ***

Iter 031: Subset=['petal-width', 'z2', 'z3', 'z4'] | Accuracy=0.9533 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 032: Subset=['petal-width', 'z1', 'z2', 'z3', 'z4'] | Accuracy=0.9400 | Pr[accept]=0.980583 | RandomUniform=0.527380 | Status=Accepted

Iter 033: Subset=['petal-width', 'z2', 'z3', 'z4'] | Accuracy=0.9533 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 034: Subset=['petal-width', 'z3'] | Accuracy=0.9400 | Pr[accept]=0.980001 | RandomUniform=0.322002 | Status=Accepted

Iter 035: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z3'] | Accuracy=0.9333 | Pr[accept]=0.989796 | RandomUniform=0.312005 | Status=Accepted

Iter 036: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z4'] | Accuracy=0.9333 | Pr[accept]=1.000000 | RandomUniform=0.188039 | Status=Accepted

Iter 037: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 038: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z3', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 039: Subset=['sepal-length', 'sepal-width', 'z1', 'z3', 'z4'] | Accuracy=0.9600 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

Iter 040: Subset=['sepal-length', 'sepal-width', 'z1', 'z4'] | Accuracy=0.9667 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved

*** Restart at iter 40: New random subset=['sepal-length', 'sepal-width', 'petal-length', 'z2', 'z4'] | Accuracy=0.9267 ***

Iter 041: Subset=['sepal-length', 'sepal-width', 'z1', 'z3', 'z4'] | Accuracy=0.9600 | Pr[accept]=0.988764 | RandomUniform=0.674903 | Status=Accepted

Iter 042: Subset=['sepal-width', 'z1', 'z4'] | Accuracy=0.8867 | Pr[accept]=0.881230 | RandomUniform=0.737451 | Status=Accepted
 Iter 043: Subset=['sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 044: Subset=['sepal-width', 'petal-width', 'z4'] | Accuracy=0.9333 | Pr[accept]=0.976472 | RandomUniform=0.070727 | Status=Accepted
 Iter 045: Subset=['sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 046: Subset=['sepal-width', 'petal-length', 'z1', 'z4'] | Accuracy=0.9267 | Pr[accept]=0.963640 | RandomUniform=0.564941 | Status=Accepted
 Iter 047: Subset=['sepal-length', 'sepal-width', 'petal-length', 'z1', 'z4'] | Accuracy=0.9533 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 048: Subset=['sepal-length', 'sepal-width', 'petal-length', 'petal-width', 'z1', 'z4'] | Accuracy=0.9467 | Pr[accept]=0.987261 | RandomUniform=0.962201 | Status=Accepted
 Iter 049: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z3', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=0.867781 | Status=Accepted
 Iter 050: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9400 | Pr[accept]=0.986755 | RandomUniform=0.200407 | Status=Accepted
 *** Restart at iter 50: New random subset=['sepal-length', 'sepal-width', 'petal-width', 'z2'] | Accuracy=0.9200 ***
 Iter 051: Subset=['sepal-width', 'petal-length', 'petal-width', 'z1', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 052: Subset=['sepal-length', 'sepal-width', 'petal-length', 'petal-width', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=0.156965 | Status=Accepted
 Iter 053: Subset=['sepal-length', 'sepal-width', 'petal-width'] | Accuracy=0.9200 | Pr[accept]=0.944842 | RandomUniform=0.429240 | Status=Accepted
 Iter 054: Subset=['sepal-length', 'petal-width', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 055: Subset=['sepal-length', 'petal-width', 'z2'] | Accuracy=0.9267 | Pr[accept]=0.970805 | RandomUniform=0.687110 | Status=Accepted
 Iter 056: Subset=['z2'] | Accuracy=0.3067 | Pr[accept]=0.244365 | RandomUniform=0.725465 | Status=Discarded

Iter 057: Subset=['sepal-length', 'petal-length', 'petal-width', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 058: Subset=['sepal-length', 'petal-width', 'z2'] | Accuracy=0.9267 | Pr[accept]=0.968753 | RandomUniform=0.910988 | Status=Accepted
 Iter 059: Subset=['sepal-length', 'petal-width', 'z1', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 060: Subset=['sepal-length', 'petal-length', 'petal-width', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.338852 | Status=Accepted
 *** Restart at iter 60: New random subset=['sepal-length', 'petal-length', 'z1', 'z2', 'z4'] | Accuracy=0.9200 ***
 Iter 061: Subset=['sepal-length', 'sepal-width', 'petal-length', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.866168 | Status=Accepted
 Iter 062: Subset=['sepal-length', 'sepal-width', 'z2'] | Accuracy=0.7067 | Pr[accept]=0.541163 | RandomUniform=0.340897 | Status=Accepted
 Iter 063: Subset=['sepal-length', 'z1', 'z2'] | Accuracy=0.9600 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 064: Subset=['sepal-length', 'sepal-width', 'z1', 'z2'] | Accuracy=0.9467 | Pr[accept]=0.963640 | RandomUniform=0.952246 | Status=Accepted
 Iter 065: Subset=['sepal-length', 'sepal-width', 'z2', 'z4'] | Accuracy=0.7333 | Pr[accept]=0.543610 | RandomUniform=0.534294 | Status=Accepted
 Iter 066: Subset=['sepal-length', 'sepal-width', 'z2'] | Accuracy=0.7067 | Pr[accept]=0.924566 | RandomUniform=0.896759 | Status=Accepted
 Iter 067: Subset=['sepal-width', 'z2'] | Accuracy=0.5467 | Pr[accept]=0.615790 | RandomUniform=0.827768 | Status=Discarded
 Iter 068: Subset=['sepal-width', 'z2'] | Accuracy=0.5467 | Pr[accept]=0.606531 | RandomUniform=0.091683 | Status=Accepted
 Iter 069: Subset=['sepal-length', 'sepal-width', 'z2'] | Accuracy=0.7067 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 070: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z2'] | Accuracy=0.9200 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 *** Restart at iter 70: New random subset=['sepal-width', 'petal-length', 'z1', 'z2', 'z3', 'z4'] | Accuracy=0.9267 ***
 Iter 071: Subset=['sepal-width', 'petal-length', 'z1', 'z2'] | Accuracy=0.9267 | Pr[accept]=1.000000 | RandomUniform=0.970264 | Status=Accepted

Iter 072: Subset=['sepal-width', 'petal-length'] | Accuracy=0.9067 | Pr[accept]=0.931063 | RandomUniform=0.477010 | Status=Accepted
 Iter 073: Subset=['sepal-width', 'petal-length', 'petal-width'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 074: Subset=['sepal-length', 'sepal-width', 'petal-length', 'petal-width', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=0.010506 | Status=Accepted
 Iter 075: Subset=['sepal-length', 'sepal-width', 'petal-length', 'z2', 'z4'] | Accuracy=0.9267 | Pr[accept]=0.923116 | RandomUniform=0.391269 | Status=Accepted
 Iter 076: Subset=['petal-length', 'z2', 'z4'] | Accuracy=0.9133 | Pr[accept]=0.945959 | RandomUniform=0.913392 | Status=Accepted
 Iter 077: Subset=['sepal-width', 'z2', 'z4'] | Accuracy=0.5733 | Pr[accept]=0.228034 | RandomUniform=0.215193 | Status=Accepted
 Iter 078: Subset=['sepal-width', 'petal-length', 'z4'] | Accuracy=0.9133 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 079: Subset=['sepal-width', 'petal-length', 'petal-width', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 080: Subset=['petal-length', 'z4'] | Accuracy=0.9267 | Pr[accept]=0.904837 | RandomUniform=0.360464 | Status=Accepted
 *** Restart at iter 80: New random subset=['sepal-width', 'petal-width', 'z2'] | Accuracy=0.9200 ***
 Iter 081: Subset=['petal-length', 'z2', 'z4'] | Accuracy=0.9133 | Pr[accept]=0.932230 | RandomUniform=0.186813 | Status=Accepted
 Iter 082: Subset=['petal-width', 'z2', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 083: Subset=['sepal-width', 'petal-width', 'z2', 'z4'] | Accuracy=0.9133 | Pr[accept]=0.854821 | RandomUniform=0.245098 | Status=Accepted
 Iter 084: Subset=['sepal-width', 'z2', 'z4'] | Accuracy=0.5733 | Pr[accept]=0.119433 | RandomUniform=0.401774 | Status=Discarded
 Iter 085: Subset=['petal-width', 'z4'] | Accuracy=0.9267 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 086: Subset=['sepal-length', 'petal-width', 'z4'] | Accuracy=0.9267 | Pr[accept]=1.000000 | RandomUniform=0.635382 | Status=Accepted
 Iter 087: Subset=['sepal-length', 'z2', 'z4'] | Accuracy=0.7667 | Pr[accept]=0.292068 | RandomUniform=0.468834 | Status=Discarded

Iter 088: Subset=['sepal-length', 'sepal-width', 'petal-width', 'z4'] | Accuracy=0.9333 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 089: Subset=['sepal-length', 'petal-width', 'z1', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 090: Subset=['sepal-length', 'petal-length', 'petal-width', 'z1', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=0.776209 | Status=Accepted
 *** Restart at iter 90: New random subset=['sepal-length', 'petal-width', 'z2', 'z3', 'z4'] | Accuracy=0.9400 ***
 Iter 091: Subset=['sepal-length', 'petal-length', 'petal-width', 'z1', 'z2', 'z4'] | Accuracy=0.9400 | Pr[accept]=0.928603 | RandomUniform=0.821030 | Status=Accepted
 Iter 092: Subset=['sepal-length', 'petal-width', 'z1', 'z2', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.298926 | Status=Accepted
 Iter 093: Subset=['petal-width', 'z1', 'z2', 'z4'] | Accuracy=0.9467 | Pr[accept]=1.000000 | RandomUniform=- | Status=Improved
 Iter 094: Subset=['petal-width', 'z2', 'z4'] | Accuracy=0.9400 | Pr[accept]=0.894839 | RandomUniform=0.777777 | Status=Accepted
 Iter 095: Subset=['petal-width', 'z1', 'z2', 'z3', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.250220 | Status=Accepted
 Iter 096: Subset=['sepal-length', 'petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.395710 | Status=Accepted
 Iter 097: Subset=['sepal-length', 'petal-length', 'petal-width', 'z1', 'z2', 'z3', 'z4'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.485783 | Status=Accepted
 Iter 098: Subset=['petal-length', 'petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.691703 | Status=Accepted
 Iter 099: Subset=['petal-width', 'z1', 'z2', 'z3'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.138496 | Status=Accepted
 Iter 100: Subset=['petal-width', 'z2'] | Accuracy=0.9400 | Pr[accept]=1.000000 | RandomUniform=0.401082 | Status=Accepted
 *** Restart at iter 100: New random subset=['sepal-length', 'sepal-width', 'petal-width', 'z1', 'z4'] | Accuracy=0.9400 ***

Simulated Annealing final selected features: ['petal-width', 'z2']

Confusion matrix:

```
[ 50  0  0 ]  
[  0 47  3 ]  
[  0  6 44 ]  
Accuracy: 0.9400
```

Part 4: Genetic Algorithm

Generation 1 top 5:

1. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, z1, z2, z3, z4] | Accuracy=0.9600
3. Features=[sepal-width, petal-width, z3, z4] | Accuracy=0.9533
4. Features=[sepal-length, sepal-width, petal-length, petal-width, z1] | Accuracy=0.9467
5. Features=[sepal-width, petal-width, z1, z4] | Accuracy=0.9467

Generation 2 top 5:

1. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, sepal-width, z1] | Accuracy=0.9600
3. Features=[sepal-width, z1, z2, z4] | Accuracy=0.9533
4. Features=[sepal-width, petal-width, z3, z4] | Accuracy=0.9533
5. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z4] | Accuracy=0.9467

Generation 3 top 5:

1. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600
2. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
3. Features=[sepal-width, z1, z2, z4] | Accuracy=0.9533
4. Features=[sepal-width, petal-length, petal-width, z1, z4] | Accuracy=0.9467
5. Features=[sepal-length, sepal-width, petal-length, petal-width, z3, z4] | Accuracy=0.9467

Generation 4 top 5:

1. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-width, z1, z3] | Accuracy=0.9600
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z3, z4]

| Accuracy=0.9467

4. Features=[sepal-length, petal-length, petal-width, z3, z4] | Accuracy=0.9467

5. Features=[sepal-width, petal-width, z1, z2, z3, z4] | Accuracy=0.9467

Generation 5 top 5:

1. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600

2. Features=[sepal-length, sepal-width, z1, z3, z4] | Accuracy=0.9600

3. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600

4. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600

5. Features=[sepal-length, sepal-width, petal-length, petal-width, z3, z4] | Accuracy=0.9467

Generation 6 top 5:

1. Features=[z1, z3, z4] | Accuracy=0.9667

2. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600

3. Features=[sepal-width, petal-length, petal-width, z2, z4] | Accuracy=0.9467

4. Features=[sepal-width, petal-length, petal-width, z3, z4] | Accuracy=0.9467

5. Features=[sepal-width, petal-length, petal-width, z3] | Accuracy=0.9467

Generation 7 top 5:

1. Features=[sepal-width, z1, z3] | Accuracy=0.9600

2. Features=[sepal-width, z1, z3] | Accuracy=0.9600

3. Features=[sepal-width, petal-width, z3, z4] | Accuracy=0.9533

4. Features=[sepal-length, sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9533

5. Features=[sepal-length, sepal-width, petal-length, z1, z4] | Accuracy=0.9533

Generation 8 top 5:

1. Features=[sepal-length, sepal-width, z1, z2, z3] | Accuracy=0.9600

2. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600

3. Features=[z1, z2, z3, z4] | Accuracy=0.9600

4. Features=[sepal-width, petal-width, z3, z4] | Accuracy=0.9533

5. Features=[sepal-width, petal-length, petal-width, z3, z4] | Accuracy=0.9467

Generation 9 top 5:

1. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600
3. Features=[sepal-length, sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9533
4. Features=[sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9467
5. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

Generation 10 top 5:

1. Features=[z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, sepal-width, z1] | Accuracy=0.9600
3. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600
4. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
5. Features=[sepal-width, z1, z2] | Accuracy=0.9467

Generation 11 top 5:

1. Features=[sepal-width, petal-length, petal-width, z2, z4] | Accuracy=0.9467
2. Features=[sepal-width, petal-length, petal-width, z3, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
4. Features=[petal-length, z3, z4] | Accuracy=0.9467
5. Features=[sepal-length, petal-length, petal-width, z1, z2, z3, z4] | Accuracy=0.9400

Generation 12 top 5:

1. Features=[sepal-length, z1, z3] | Accuracy=0.9600
2. Features=[sepal-width, z1, z2, z4] | Accuracy=0.9533
3. Features=[sepal-width, petal-length, petal-width, z2] | Accuracy=0.9467
4. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

9467

5. Features=[sepal-width, petal-length, petal-width, z2, z4] | Accuracy=0.9467

Generation 13 top 5:

1. Features=[sepal-width, z1, z3] | Accuracy=0.9600

2. Features=[sepal-length, z1, z3] | Accuracy=0.9600

3. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

4. Features=[sepal-width, petal-length, petal-width, z2] | Accuracy=0.9467

5. Features=[sepal-length, sepal-width, petal-length, petal-width, z1] | Accuracy=0.9467

Generation 14 top 5:

1. Features=[sepal-length, z1, z3] | Accuracy=0.9600

2. Features=[sepal-length, z1, z3] | Accuracy=0.9600

3. Features=[sepal-length, sepal-width, petal-length, petal-width] | Accuracy=0.9467

4. Features=[petal-length, petal-width, z1, z3] | Accuracy=0.9467

5. Features=[sepal-width, petal-length, petal-width, z1] | Accuracy=0.9467

Generation 15 top 5:

1. Features=[sepal-length, z1] | Accuracy=0.9667

2. Features=[sepal-length, z1] | Accuracy=0.9667

3. Features=[sepal-width, petal-length, petal-width, z2, z4] | Accuracy=0.9467

4. Features=[sepal-length, sepal-width, petal-length, z3] | Accuracy=0.9467

5. Features=[sepal-length, sepal-width, petal-width, z1, z3] | Accuracy=0.9400

Generation 16 top 5:

1. Features=[sepal-length, z1, z3] | Accuracy=0.9600

2. Features=[petal-width] | Accuracy=0.9533

3. Features=[sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467

4. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z3] | A

ccuracy=0.9467

5. Features=[sepal-length, sepal-width, petal-width, z1, z3, z4] | Accuracy=0.9467

Generation 17 top 5:

1. Features=[sepal-length, z1, z3, z4] | Accuracy=0.9600

2. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

3. Features=[sepal-width, petal-width, z1, z3, z4] | Accuracy=0.9467

4. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

5. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

Generation 18 top 5:

1. Features=[petal-width, z2, z3, z4] | Accuracy=0.9533

2. Features=[petal-length, petal-width, z1, z3, z4] | Accuracy=0.9467

3. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

4. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

5. Features=[sepal-length, sepal-width, petal-length, z1, z3, z4] | Accuracy=0.9467

Generation 19 top 5:

1. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

2. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

3. Features=[petal-width, z3, z4] | Accuracy=0.9467

4. Features=[petal-length, petal-width, z1] | Accuracy=0.9467

5. Features=[sepal-length, sepal-width, petal-length, petal-width] | Accuracy=0.9467

Generation 20 top 5:

1. Features=[sepal-length, z1, z2, z3, z4] | Accuracy=0.9600

2. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3, z4] | Accuracy=0.9467
4. Features=[sepal-length, sepal-width, petal-length, z3] | Accuracy=0.9467
5. Features=[petal-length, z1] | Accuracy=0.9467

Generation 21 top 5:

1. Features=[sepal-width, petal-length, petal-width, z3] | Accuracy=0.9467
2. Features=[sepal-length, sepal-width, petal-length, petal-width, z1] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
4. Features=[sepal-length, petal-length, z1] | Accuracy=0.9400
5. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2] | Accuracy=0.9400

Generation 22 top 5:

1. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
2. Features=[petal-length, petal-width, z1, z3] | Accuracy=0.9467
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
4. Features=[sepal-width, z1, z2] | Accuracy=0.9467
5. Features=[petal-length, petal-width, z1, z2] | Accuracy=0.9400

Generation 23 top 5:

1. Features=[sepal-length, petal-length, z4] | Accuracy=0.9533
2. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
4. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
5. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

Generation 24 top 5:

1. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
2. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
3. Features=[sepal-length, sepal-width, petal-length, z1, z2, z3] | Accuracy=0.9467
4. Features=[z1, z2, z3] | Accuracy=0.9467
5. Features=[sepal-length, petal-length, petal-width, z3] | Accuracy=0.9400

Generation 25 top 5:

1. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
2. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
4. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
5. Features=[sepal-length, sepal-width, petal-length, z2] | Accuracy=0.9400

Generation 26 top 5:

1. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
2. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467
4. Features=[sepal-length, sepal-width, z1, z2] | Accuracy=0.9467
5. Features=[sepal-length, petal-length, z1] | Accuracy=0.9400

Generation 27 top 5:

1. Features=[sepal-width, petal-length, petal-width, z1] | Accuracy=0.9467
2. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467

3. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467

4. Features=[sepal-width, petal-width, z1, z2, z3, z4] | Accuracy=0.9467

5. Features=[sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9467

Generation 28 top 5:

1. Features=[sepal-width, petal-length, petal-width, z4] | Accuracy=0.9467

2. Features=[sepal-width, petal-length, petal-width, z2] | Accuracy=0.9467

3. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

4. Features=[sepal-width, z1, z2] | Accuracy=0.9467

5. Features=[sepal-width, petal-width, z1, z2] | Accuracy=0.9400

Generation 29 top 5:

1. Features=[petal-width, z2, z3, z4] | Accuracy=0.9533

2. Features=[sepal-width, petal-width, z1, z3, z4] | Accuracy=0.9467

3. Features=[petal-length, petal-width, z4] | Accuracy=0.9467

4. Features=[sepal-length, petal-width, z1, z4] | Accuracy=0.9467

5. Features=[sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9467

Generation 30 top 5:

1. Features=[petal-width, z2, z3, z4] | Accuracy=0.9533

2. Features=[petal-width] | Accuracy=0.9533

3. Features=[petal-length, petal-width, z1, z3] | Accuracy=0.9467

4. Features=[sepal-width, petal-width, z1, z4] | Accuracy=0.9467

5. Features=[sepal-width, petal-length, petal-width, z1] | Accuracy=0.9467

Generation 31 top 5:

1. Features=[sepal-length, z1, z3, z4] | Accuracy=0.9600

2. Features=[petal-width, z1, z4] | Accuracy=0.9467

3. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

4. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467

5. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

Generation 32 top 5:

1. Features=[sepal-length, z1, z3] | Accuracy=0.9600
2. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
4. Features=[sepal-width, petal-length, petal-width, z3] | Accuracy=0.9467
5. Features=[sepal-width, petal-length, petal-width, z1, z2, z3, z4] | Accuracy=0.9400

Generation 33 top 5:

1. Features=[sepal-width, petal-length, petal-width] | Accuracy=0.9467
2. Features=[sepal-width, petal-width, z1, z3, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467
4. Features=[sepal-length, petal-length, z1, z2] | Accuracy=0.9400
5. Features=[sepal-width, petal-length, z1, z3] | Accuracy=0.9400

Generation 34 top 5:

1. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-width, petal-length, petal-width, z1, z4] | Accuracy=0.9467
3. Features=[petal-length, petal-width, z1, z3, z4] | Accuracy=0.9467
4. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
5. Features=[petal-length, petal-width, z2, z3, z4] | Accuracy=0.9400

Generation 35 top 5:

1. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, z1, z3] | Accuracy=0.9600
3. Features=[sepal-width, z1, z2, z4] | Accuracy=0.9533
4. Features=[sepal-width, petal-length, petal-width, z1, z2, z3] | Accuracy=0.9467
5. Features=[sepal-width, petal-length, petal-width, z1, z4] | Accuracy=0.9467

Generation 36 top 5:

1. Features=[sepal-length, petal-width, z1, z4] | Accuracy=0.9467
2. Features=[petal-length, petal-width, z1] | Accuracy=0.9467
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z4] | Accuracy=0.9467
4. Features=[sepal-length, sepal-width, petal-length, petal-width] | Accuracy=0.9467
5. Features=[petal-length, petal-width, z4] | Accuracy=0.9467

Generation 37 top 5:

1. Features=[sepal-length, petal-width, z3, z4] | Accuracy=0.9533
2. Features=[petal-length, petal-width, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z3] | Accuracy=0.9467
4. Features=[petal-length, petal-width, z4] | Accuracy=0.9467
5. Features=[petal-length, petal-width, z1] | Accuracy=0.9467

Generation 38 top 5:

1. Features=[petal-width, z2, z3, z4] | Accuracy=0.9533
2. Features=[sepal-width, petal-length, petal-width, z1, z4] | Accuracy=0.9467
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467
4. Features=[sepal-length, petal-length, petal-width, z1, z4] | Accuracy=0.9467
5. Features=[petal-length, petal-width, z1, z3] | Accuracy=0.9467

Generation 39 top 5:

1. Features=[z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[petal-width, z2, z3, z4] | Accuracy=0.9533
3. Features=[sepal-length, sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9533
4. Features=[petal-length, petal-width, z1, z3] | Accuracy=0.9467
5. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467

Generation 40 top 5:

1. Features=[petal-width] | Accuracy=0.9533
2. Features=[petal-length, petal-width, z1, z3] | Accuracy=0.9467
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z2, z3, z4] | Accuracy=0.9467
4. Features=[sepal-length, sepal-width, petal-length, z2, z3] | Accuracy=0.9400
5. Features=[sepal-length, petal-length, petal-width, z1, z3] | Accuracy=0.9400

Generation 41 top 5:

1. Features=[petal-width] | Accuracy=0.9533
2. Features=[sepal-width, petal-width, z1, z2, z3, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9467
4. Features=[petal-length, petal-width, z1] | Accuracy=0.9467
5. Features=[sepal-length, sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467

Generation 42 top 5:

1. Features=[sepal-length, sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, sepal-width, z1, z2, z3] | Accuracy=0.9600
3. Features=[sepal-length, z1, z3] | Accuracy=0.9600
4. Features=[sepal-length, sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9533
5. Features=[sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9467

Generation 43 top 5:

1. Features=[sepal-length, sepal-width, z1, z2, z3] | Accuracy=0.9600
2. Features=[sepal-length, sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
3. Features=[sepal-length, sepal-width, petal-length, petal-width, z3] | Accuracy=0.9467
4. Features=[sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467
5. Features=[sepal-width, petal-width, z1, z2, z3, z4] | Accuracy=0.9467

Generation 44 top 5:

1. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600
2. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
3. Features=[sepal-length, z1, z2] | Accuracy=0.9600
4. Features=[sepal-length, sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
5. Features=[sepal-length, sepal-width, z1, z2, z3] | Accuracy=0.9600

Generation 45 top 5:

1. Features=[sepal-length, sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, sepal-width, petal-width, z1, z3, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467
4. Features=[sepal-length, sepal-width, petal-length, z3] | Accuracy=0.9467
5. Features=[z1, z2] | Accuracy=0.9467

Generation 46 top 5:

1. Features=[sepal-width, z1, z3, z4] | Accuracy=0.9600
2. Features=[sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
3. Features=[sepal-length, sepal-width, petal-width, z3, z4] | Accuracy=0.9533
4. Features=[sepal-length, sepal-width, petal-width, z1, z3, z4] | Accuracy=0.9467
5. Features=[z1, z2, z3] | Accuracy=0.9467

Generation 47 top 5:

1. Features=[sepal-length, z1] | Accuracy=0.9667
2. Features=[sepal-length, sepal-width, petal-length, petal-width, z3, z4] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z2, z3] | Accuracy=0.9467
4. Features=[z1, z2] | Accuracy=0.9467
5. Features=[petal-length, z3] | Accuracy=0.9467

Generation 48 top 5:

1. Features=[sepal-width, petal-length, petal-width, z2] | Accuracy=0.9467
2. Features=[sepal-width, petal-length, petal-width, z2] | Accuracy=0.9467

3. Features=[petal-width, z3, z4] | Accuracy=0.9467
4. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467
5. Features=[sepal-length, petal-width, z1, z2, z3, z4] | Accuracy=0.9467

Generation 49 top 5:

1. Features=[sepal-length, z1, z2, z4] | Accuracy=0.9533
2. Features=[sepal-width, petal-length, petal-width, z1, z3] | Accuracy=0.9467
3. Features=[sepal-width, petal-length, petal-width, z4] | Accuracy=0.9467
4. Features=[petal-width, z3, z4] | Accuracy=0.9467
5. Features=[sepal-width, petal-length, petal-width, z1, z3, z4] | Accuracy=0.9467

Generation 50 top 5:

1. Features=[sepal-length, sepal-width, z1, z2, z3, z4] | Accuracy=0.9600
2. Features=[sepal-length, sepal-width, petal-width, z2, z3, z4] | Accuracy=0.9533
3. Features=[sepal-length, sepal-width, petal-width, z3, z4] | Accuracy=0.9533
4. Features=[petal-width] | Accuracy=0.9533
5. Features=[sepal-length, sepal-width, petal-length, petal-width, z1, z2, z3, z4] | Accuracy=0.9467

Genetic Algorithm final selected features: ['sepal-length', 'sepal-width', 'z1', 'z4']

Confusion matrix:

[50 0 0]

[0 47 3]

[0 2 48]

Accuracy: 0.9667

Final Summary

Part 1 Accuracy (original 4 features): 0.9467

Part 2 Accuracy (PCA-selected features): 0.9133

Part 3 Accuracy (Simulated Annealing selected): 0.9400

Part 4 Accuracy (Genetic Algorithm selected): 0.9667

Final feature sets used:

Part 1: ['sepal-length', 'sepal-width', 'petal-length', 'petal-width']

Part 2: ['z1']

Part 3: ['petal-width', 'z2']

Part 4: ['sepal-length', 'sepal-width', 'z1', 'z4']